

Computer-aided drug design (CADD)

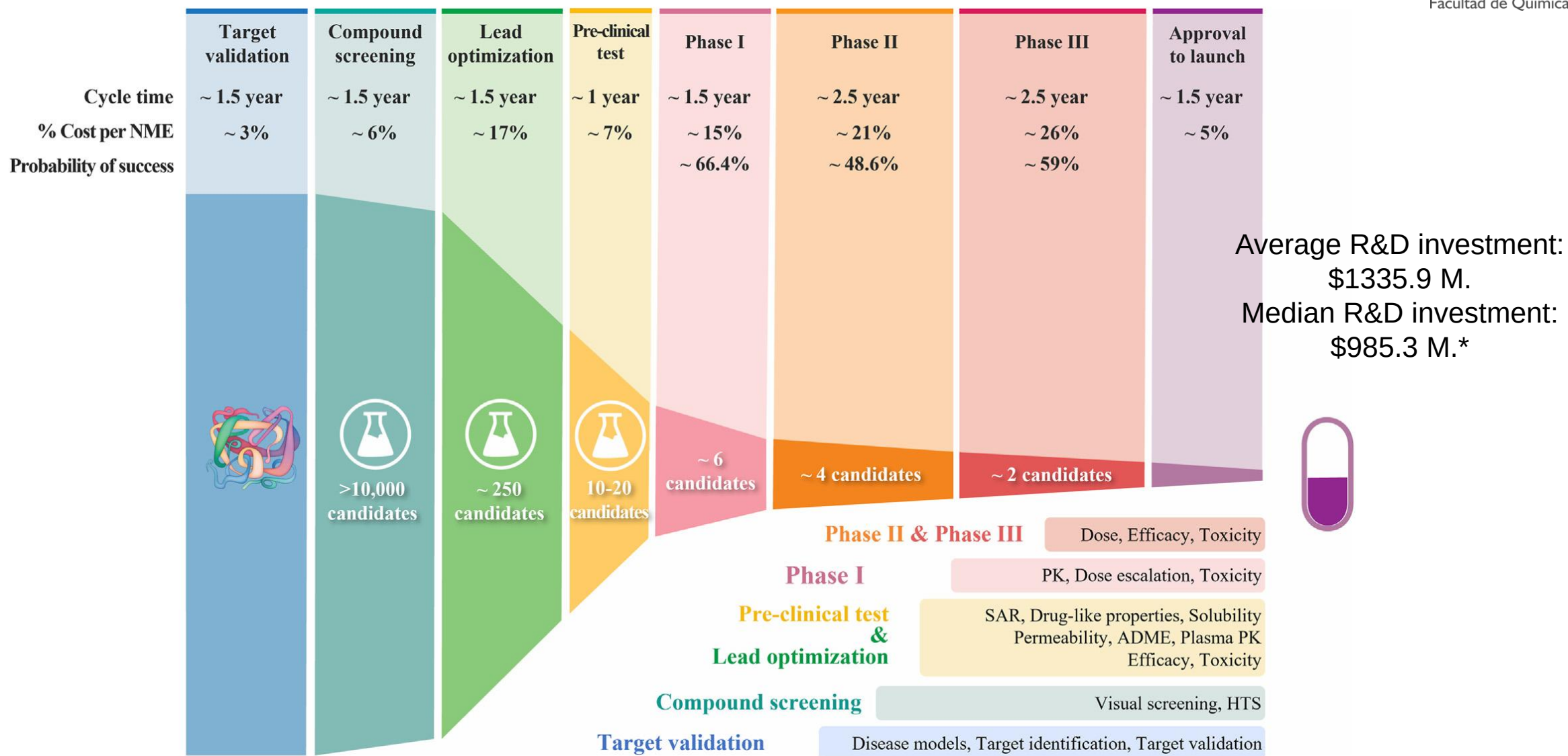
Juan David Sánchez Ramírez

Professor
Dr. José Luis Medina Franco



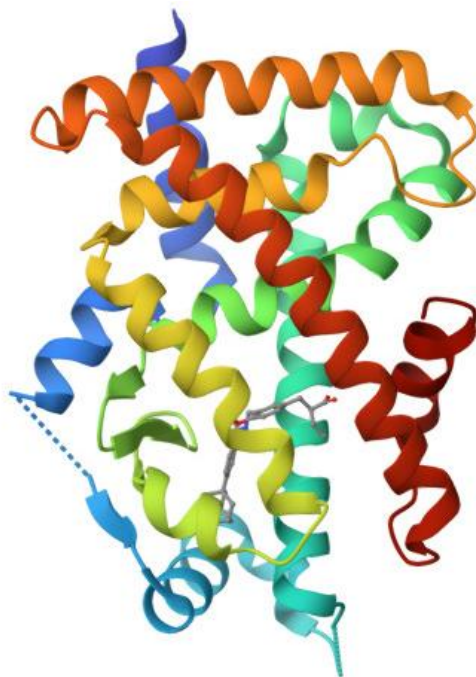
Facultad de Química
Universidad Nacional Autónoma de México

From Traditional to CADD-driven pipelines



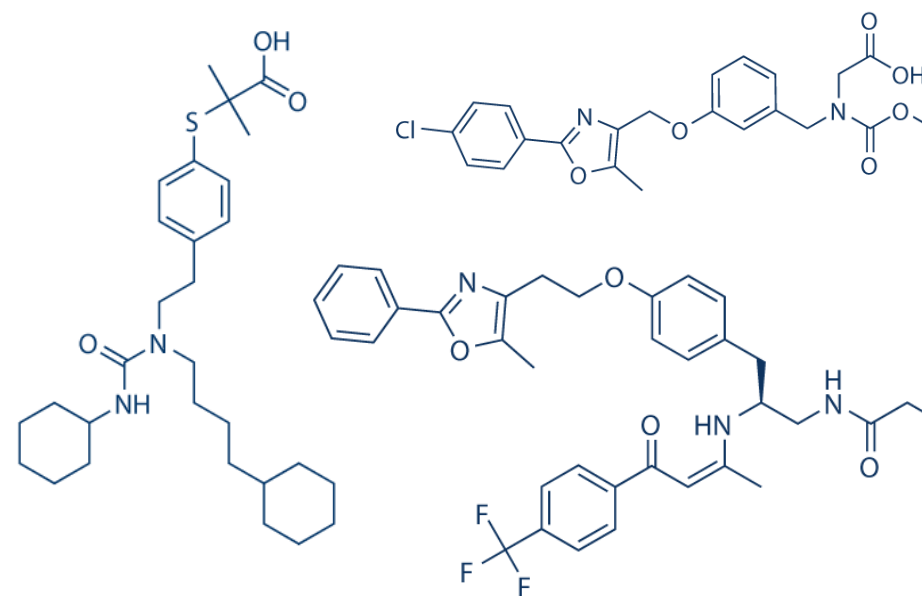
Computer-aided drug design (CADD) methodologies

Structure-based drug design (SBDD)



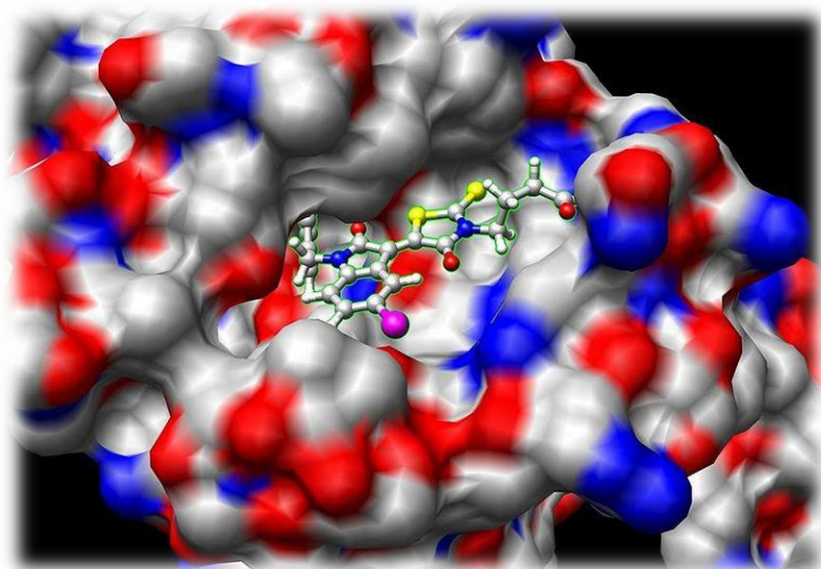
PPAR- α

Ligand-based drug design (LBDD)

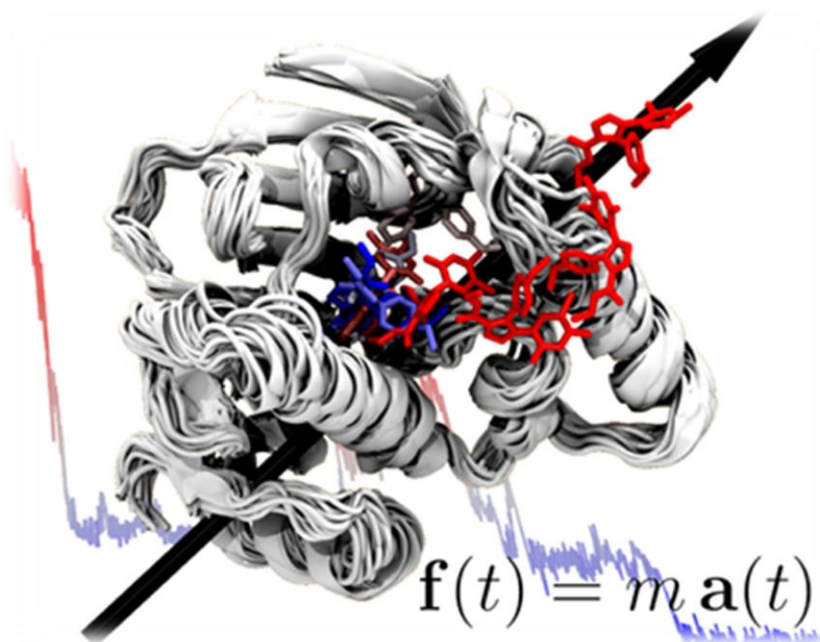


PPAR- α inhibitors

Structure-based drug design (SBDD)



Molecular docking

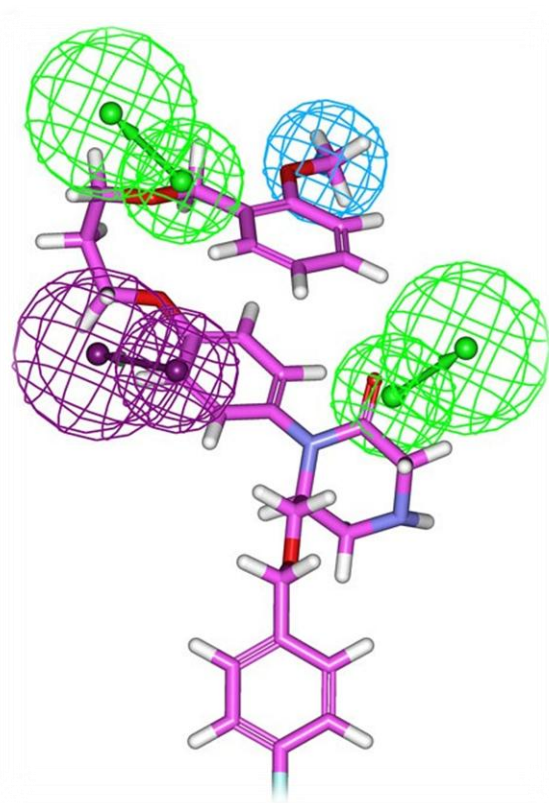


Molecular dynamics

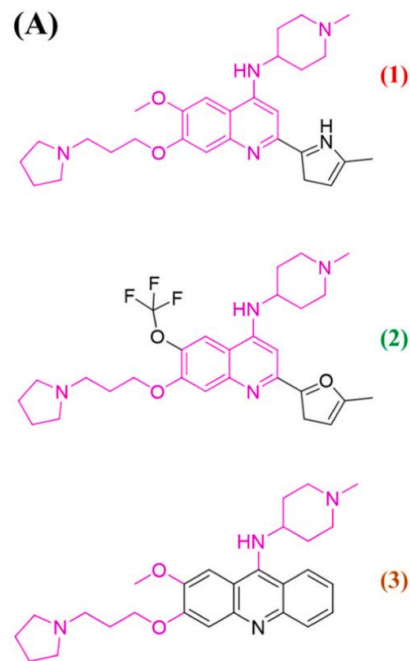
Software:

AutoDock, Glide,
Gold, Vina, MOE.
GROMACS, AMBER.

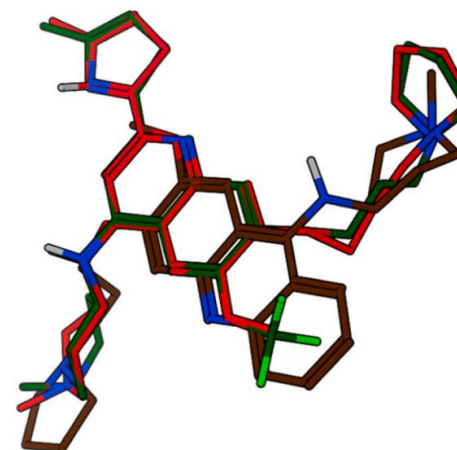
Ligand-based drug design (LBDD)



Pharmacophore modeling



Molecular similarity



Software:

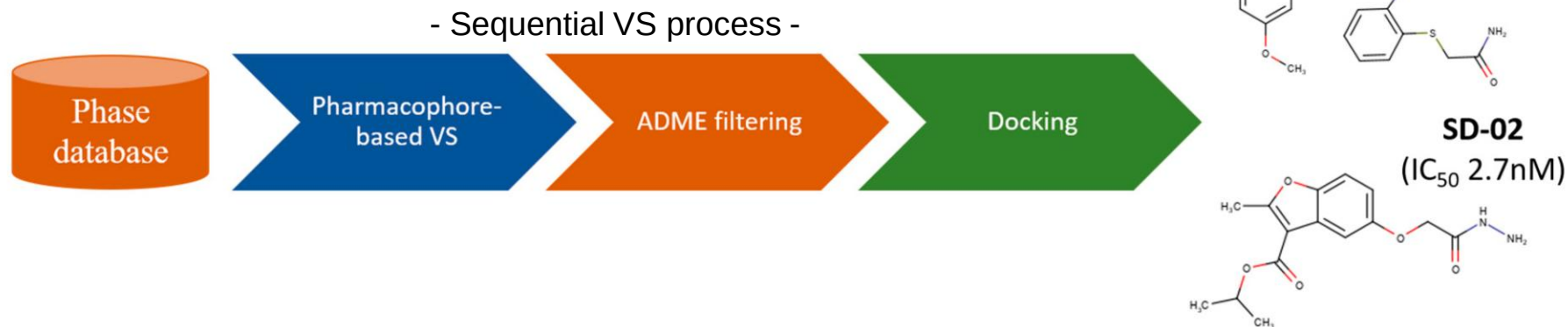
Schrödinger Phase,
Pharmer, MOE.
RDKit, SwissSimilarity.

Hybrid approaches

A.



B.

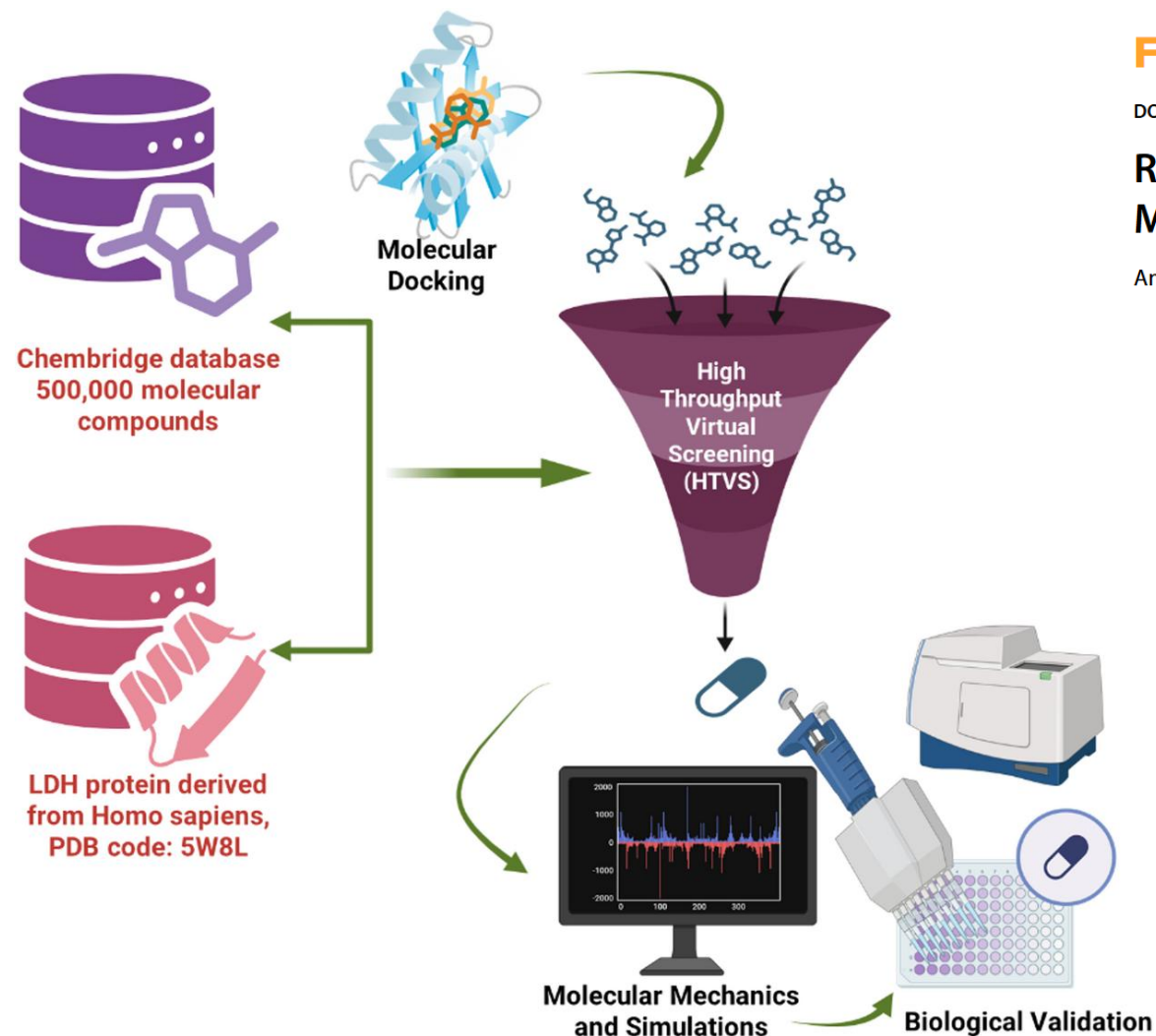


 SBDD Method

 LBDD Method

Representative cases of two combined ligand-based (LB) and structure-based (SB) strategies leading to the discovery of potent inhibitors of (A) 17 β -hydroxysteroid dehydrogenase type 1 (17 β -HSD1) and (B) histone deacetylase 8 (HDAC8) enzymes.

High throughput virtual screening (HTVS)



Full Paper

www.molinf.com

DOI: 10.1002/minf.202000028

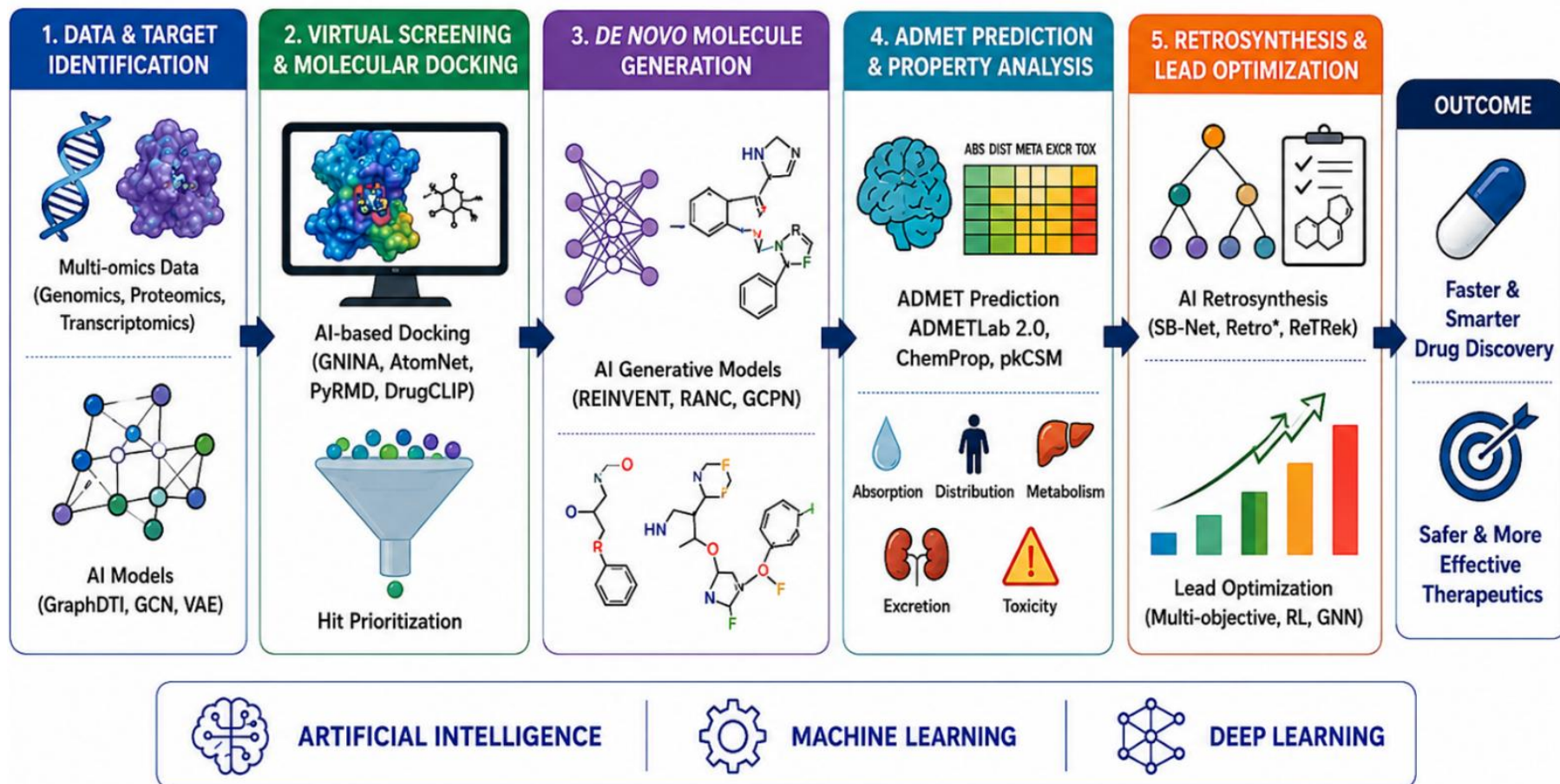
Rapid Identification of Potential Inhibitors of SARS-CoV-2 Main Protease by Deep Docking of 1.3 Billion Compounds

Anh-Tien Ton^{+, [a]} Francesco Gentile^{+, [a]} Michael Hsing^{+, [a]} Fuqiang Ban^{+, [a]} and Artem Cherkasov^{*, [a]}

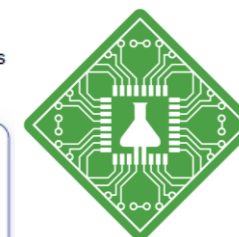
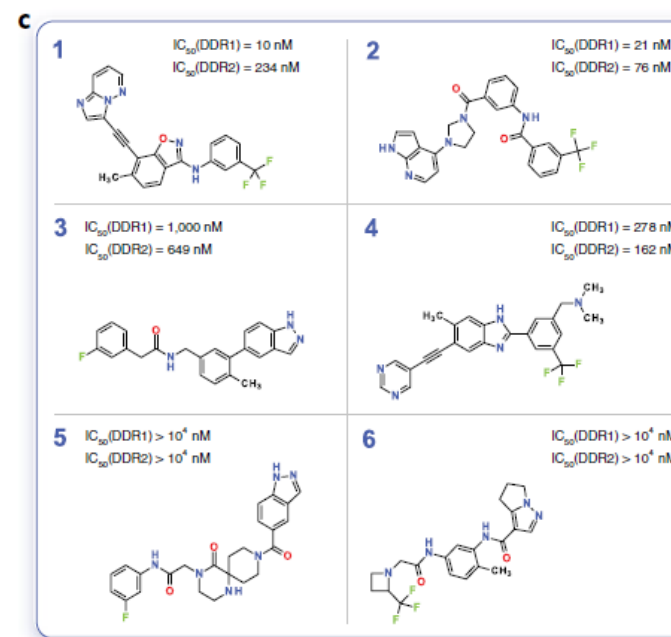
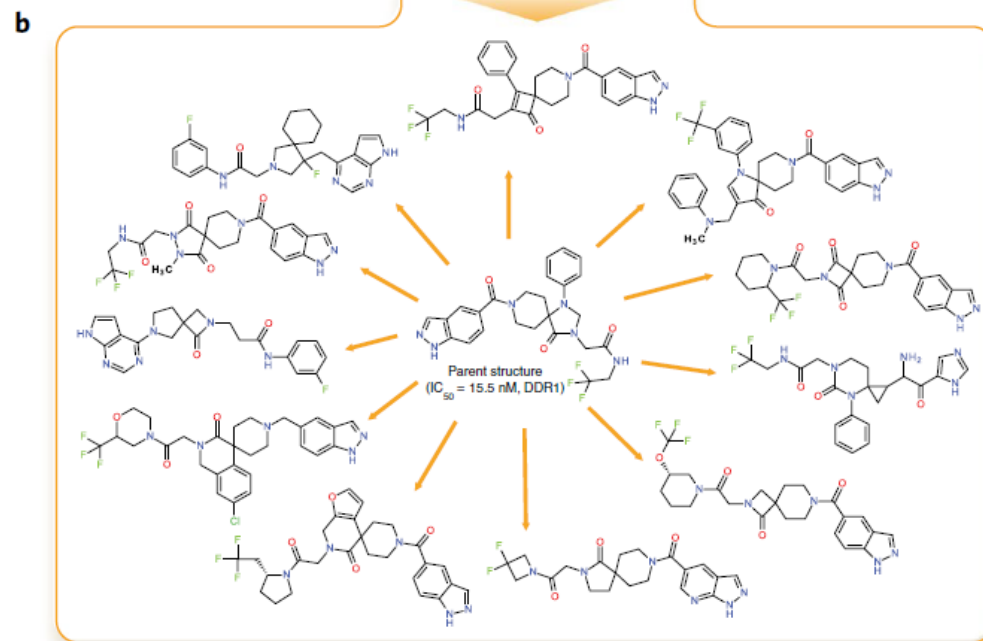
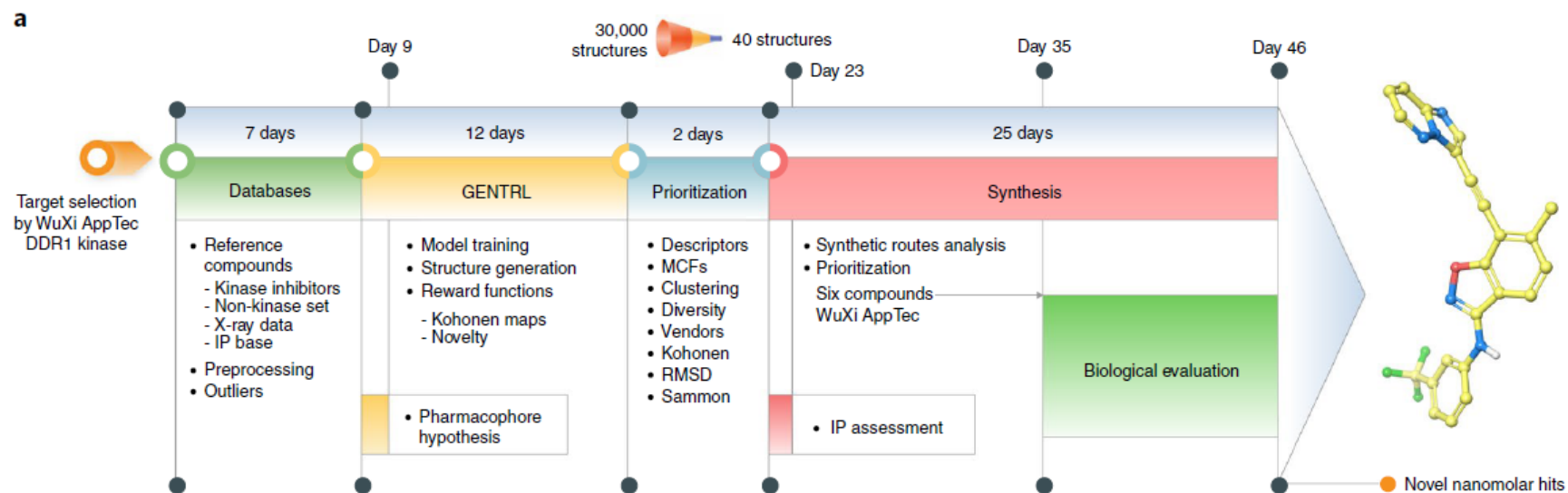


AI-DRIVEN DRUG DISCOVERY

Tools, Integrated Workflow, and Current Challenges



Frontiers in CADD



Insilico
Medicine

Frontiers in CADD

Generative AI Software for Drug Discovery,
Scientific Research & Sustainability

**SOFTWARE
FOR SCIENCE**

Insilico Medicine
insilico.com

**THERAPEUTIC
PIPELINE**

High Quality Therapeutic Programs
Discovered Using Generative AI and Automation

Biology⁴²

PandaOmics
Discover and prioritize novel targets

Generative Biologics
Discover and optimize novel biomolecules

Life Star 2
Automated lab operating environment

Large Language of Life Models (LLLMs)

Precious1GPT
Multimomics age prediction & target ID

Precious2GPT
Multimodal multimomics biological data synthesis

Precious3GPT
Multi tissue multispecies multimomics multimodal life model

Chemistry⁴²

Generative Chemistry
Generate novel molecules

Retrosynthesis
Predict synthetic routes for molecular structures

Model Training
Custom models based on in-house data

MDFlow
End-to-end simulation workflows for biomolecules and complexes

Alchemy
Physics-based relative binding free energy engine

ADMET & Off-target
Profiling & optimization

MolSpace
Visualize and analyze chemical data using dimensionality reduction technique

Nach01
Multimodal natural & chemical languages foundation model

Medicine⁴²

inClinico
Design and predict clinical trials

Science⁴²

DORA
Multi-agent generative research assistant

LLM Assistant

Copilot
Generative conversational agent

Data Warehouse
Seamless cross-application data flow via efficient integration & standardization

Environmental Sustainability
Generative AI technologies for environmental sustainability

Science MMAI Gym
Boost the LLM's intelligence in drug discovery and development

INDICATION	TARGET ID	HIT TO LEAD	LEAD OPT.	IND-ENABLING	PHASE I	PHASE II	PHASE III
Idiopathic Pulmonary Fibrosis	TNIK				China	China NMPA	
Idiopathic Pulmonary Fibrosis	TNIK				New Zealand	US FDA	
IPF Inhalable	TNIK						IND clearance
BRCA-mutated Cancer	USP1						Out-licensed to Exelixis
Immuno-Oncology	QPCTL						
ER+HER2- Breast Cancer	KAT6						Out-licensed to Menarini
Solid Tumors	KIF18A						Out-licensed to Menarini
Inflammatory Bowel Disease	PHD1/2	Gut-Restricted					
Anemia of Chronic Kidney Disease	PHD1/2						Greater China Rights Out-licensed to TaiGen
MTAP-/- Cancer	MAT2A						
Solid Tumors	TEAD						
Solid Tumors	ENPP1						IND clearance
CNS	NLRP3						Co-development with Hygia Therapeutics
Inflammatory Diseases	NLRP3	Peripheral restricted					
Obesity & Metabolic Diseases	GLP-1R						Out-licensed to undisclosed partner
Pain	Nav1.8						Greater China rights out-licensed to undisclosed partner
Pain	Target Z						
Immuno-Oncology	CBLB						
Solid Tumors	PRMT5						
Obesity & Metabolic Diseases	GLPR						
KRAS-driven Solid Tumors	Pan-KRAS						
Metabolic Disease and Oncology	NR3C1						
Metabolic Diseases	Lp(a)						
Inflammatory Diseases	VAV1						
Obesity & Metabolic Diseases	APJ						

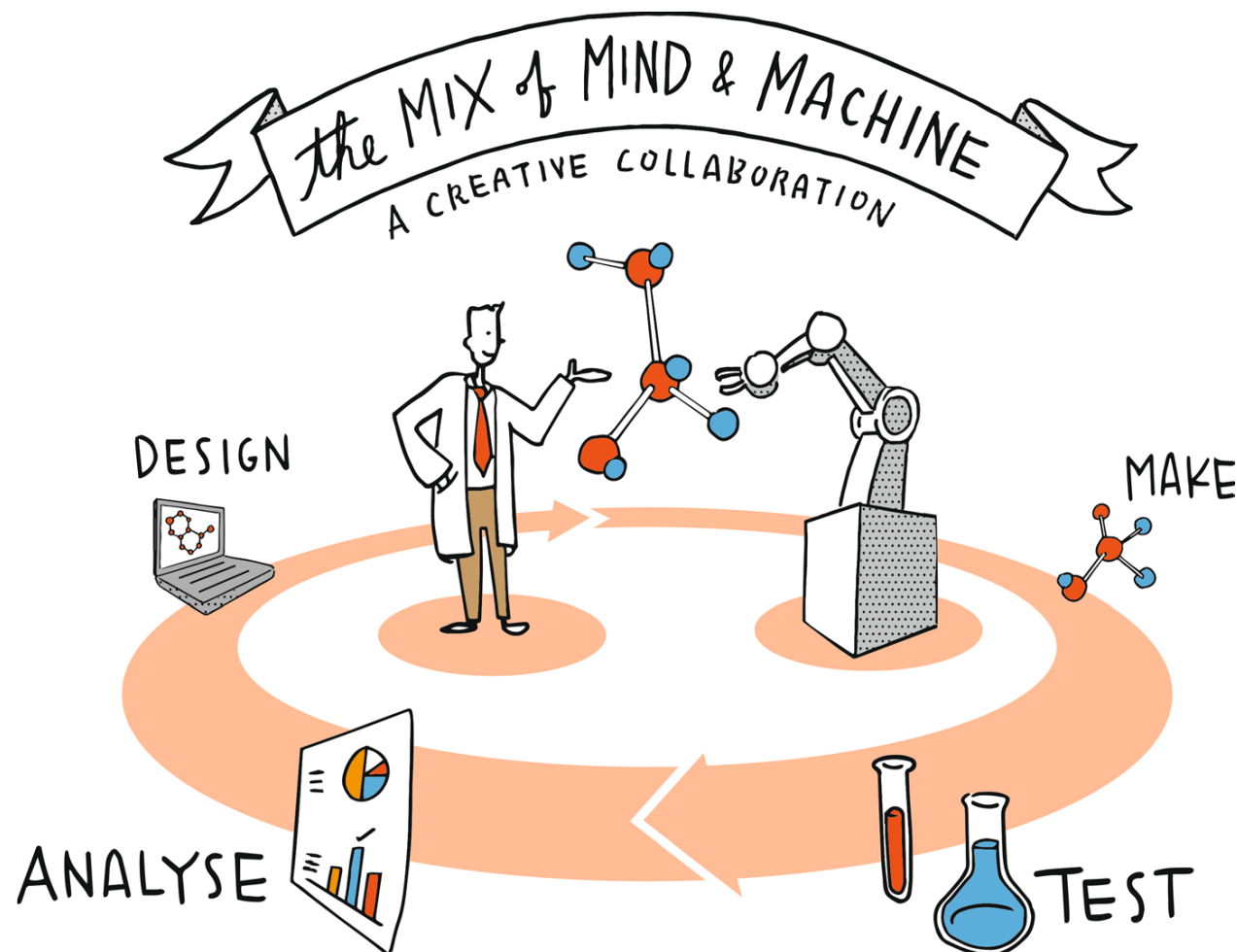
VALUES

- PATIENT FIRST
- RELENTLESS INNOVATION
- TRANSPARENCY & INTEGRITY
- HIGHEST QUALITY & REPRODUCIBILITY

Available for Licensing
More programs in discovery



Reflections



Rethinking drug design in the artificial intelligence era

Petra Schneider, W. Patrick Walters, Alleyn T. Plowright, Norman Sieroka, Jennifer Listgarten^{ID}, Robert A. Goodnow Jr., Jasmin Fisher^{ID}, Johanna M. Jansen, José S. Duca, Thomas S. Rush, Matthias Zentgraf, John Edward Hill, Elizabeth Krutoholow, Matthias Kohler, Jeff Blaney, Kimito Funatsu, Chris Luebke and Gisbert Schneider^{ID}

The Jungle of Generative Drug Discovery: Traps, Treasures, and Ways Out

Rıza Özçelik^{1,2} Francesca Grisoni^{*1,2}

What is the plausibility that all drugs will be designed by computers by the end of the decade?

José L. Medina-Franco & Edgar López-López

- Schneider, P., Walters, W.P., Plowright, A.T. et al. Rethinking drug design in the artificial intelligence era. Nat Rev Drug Discov 19, 353–364 (2020).
- Rıza Özçelik, Francesca Grisoni. The Jungle of Generative Drug Discovery: Traps, Treasures, and Ways Out, arXiv:2501.05457v1 [q-bio.BM] 24 Dec 2024.
- Medina-Franco, J. L., & López-López, E. (2024). What is the plausibility that all drugs will be designed by computers by the end of the decade? Expert Opinion on Drug Discovery, 19(5), 507–510.

